

Control Everything with Your Hand



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We all have heard about legends in every religion and culture across the world, supernatural humans who have the power to control things by simply moving their hands in the air. Such an all-mighty life is not far away, proves a Bengaluru based start-up, Tomar Technologies Pvt Ltd.

Prime, a wristband, is one of these new-generation products that give similar power to its users. It has the potential to let you control almost all electronic devices around you with simple hand gestures. On wearing this band, the user can increase the speed of a fan by rotating the hand clockwise, or reduce the television volume by lowering the hand with palms facing downwards. This band can continuously monitor the user's health too. The LCD screen on the band lets the person control each device's function when he or she points the hand towards it.

The story behind

"The major reason why we came up with such an innovation is because of my sister-in-law's mother," says Hemant Singh

Tomar, co-founder of Tomar Technologies. "She was suffering from pain in knees and it was very difficult for her to walk." This is when they had the idea of controlling a device or process without physically touching it.

"Imagine you do not have to use the remote control," adds Rohit Singh, co-founder of Tomar Technologies, "and you can control all devices by pointing towards these and using hand gestures." Another situation where Prime can be helpful is when you enter the room at night and cannot find the switchboard or remember the exact location of the switch.

Magic trick revealed

The technology used here is based on the theory called the disturbance theory. All

types of activities inside a closed space cause a disturbance inside that space.

Prime considers any system as a 3D model with X, Y and Z coordinates. Hand movements happening in this model are captured and converted into their 3D co-ordinates. Prime's intelligent network pre-trains the set of coordinates corresponding to each gesture and starts listening to the gestures it is trained for. When coordinates of the user's gesture match any of the network's gestures, it is identified and an equivalent function is performed. The system learns user patterns and adapts accordingly to create the most accurate results.

The second part is the indoor positioning system (IPS), which uses several technologies other than satellite in order to locate an object or person inside a closed space. A global positioning system (GPS) works well outside a building, but fails when you have to identify the exact location of a device inside a building.

Team Tomar is working on a coordinate-location mechanism that is much localised and can work inside any building. When the automation system is being installed, coordinates of all devices to be automated are located beforehand. This makes identification of devices easier when pointing one's hand towards it.

Other similar devices

There are multiple wearables, gesture-recognition based products available in the prototype form today. Fitbit is a wearable band, concentrating on the health-monitoring sector. Myo, the gesture-controlled armband by Thalmic Labs, is a band that allows the user to control gaming, smartphones, smarthomes, presentations and other such activities by tracking muscle movement. Reemo is another device in prototype form that serves as direct competition to Prime.

An Indian based start-up has come up with Fin, a ring that lets you control

